Matt Hall, PhD

London, UK

mattdanhall.com
mattdanhall@duck.com

SUMMARY

Passionate and highly driven Software Engineer with a deep curiosity for solving complex architectural challenges. Drawing from a PhD in Automatic Control for Robotic Systems, I bring a highly analytical mindset to designing, building, and operating resilient distributed systems in Go, Python, AWS and beyond. I thrive in environments that demand continuous learning; whether that be leading technical migrations, architecting developer platforms, or mentoring peers. I'm also a fan of a day spent gaming (board or video) or on a long hike with a great pub at the end.

SKILLS

Languages: Go, Python, Ruby, Scala, Java, TypeScript, C/C++, MATLAB

Distributed Systems: Kafka, Kafka Streams, DynamoDB, gRPC, Modular Robotic Systems

Cloud & DevOps: AWS, Terraform, Docker, K8s, Git, CI/CD, GitHub Actions, Datadog

Platform Engineering: Backend Services, APIs, Developer Tooling, Observability

Certification: Confluent Certified Developer for Apache Kafka (CCDAK), 2026

EDUCATION

The University of Sheffield **2017 - 2022**
PhD - Automatic Control and Systems Engineering
Thesis: "Active Subtraction: A Viable Method of Self-Reconfiguration for Modular Robotic Systems"

Lancaster University **2012 - 2016**
MEng - Electrical and Electronics Engineering
First Class Honors

PUBLICATIONS

- M. D. Hall et al., "Self-Reconfiguration in Two-Dimensions via Active Subtraction with Modular Robots", *Proc. RSS XVI*, 2020.
- A. Özdemir et al., "Spatial Coverage Without Computation", *IEEE ICRA*, 2019.

PROJECTS

Game Development: Build small games in time-constrained game jams using the Godot engine. Bots ranked #68/817 for presentation in Big Mode Game Jam 2025.

Web Development: Personal website built using the Hugo framework. Holiday let site built using HTML, CSS and JS.

EXPERIENCE

Deliveroo

Software Engineer *Nov 2024 - Present, London, UK*

- Architected and took full ownership of a data transformation and retention platform streaming Kafka events into DynamoDB, including for real-time rider tracking, processing ~1,000 messages per second at 99.99% uptime. Successfully consolidated infrastructure into a reusable Terraform product, eliminating ~30,000 lines of duplicated configuration.
- Led technical discovery and migration planning across 36 repositories, defining the technical strategy for retiring legacy infrastructure and transitioning to a modern, resilient sharded architecture.
- Drove the development of core services for Deliveroo and DoorDash's internal AI platform, enabling agents to securely discover and interact with systems via authenticated tool integrations.
- Maintained Kafka infrastructure used across customer, merchant and rider systems, supporting event streams processing millions of messages per second.
- Acted as a key responder on the on-call rotation for Kafka and AI platform services, rapidly diagnosing and resolving incidents across application, Kubernetes, and AWS infrastructure layers.
- Pioneered evaluation pipelines, retrieval-augmented generation (RAG) systems, and model workflows for internal AI products.
- Developed shared Kafka producer and consumer libraries for Go, Ruby, and Scala, successfully standardising event-driven integrations company-wide; used TLA+ to verify message-delivery guarantees.
- Championed system observability by building bespoke Datadog dashboards and operational tooling, significantly reducing support overhead and improving platform visibility.

BJSS

Software Engineer *Jun 2022 - Nov 2024, London, UK*

- Played a pivotal role in the critical migration of 330 million cervical screening records from regional databases into a unified AWS-based NHS Cervical Screening Management System, ensuring reliability for a life-saving national service.
- Entrusted as a core member of the go-live transition team, managing production deployments, rapid incident resolution, and complex release activities during high-pressure delivery windows.
- Engineered and maintained robust cloud infrastructure, deployment pipelines, and CI/CD tooling using Python and Terraform.
- Directed large-scale data migration, validation, and custom load/stress testing initiatives, designing DynamoDB scaling and retry solutions to guarantee data accuracy, security, and strict regulatory compliance.

The University of Sheffield

Postgraduate Researcher *Sep 2017 - Oct 2022, Sheffield, UK*

- Directed independent research, developing and evaluating novel control algorithms for autonomous modular robotic systems.
- Architected simulation and analysis tools in Python to model complex system behaviours and rigorously validate algorithm performance.
- Mentored and led laboratory sessions for engineering students, breaking down complex control systems and software principles.
- Published original research in top-tier robotics venues and served as a peer reviewer for IROS, RSS, and ICRA.